

ARXIV 2504.05118

VAPO

Value-Model-Based RL for Advanced Reasoning

AIME 2024 · avg@32

60.4

+10 vs DAPO · +13 vs DeepSeek-R1-Zero

Base model: Qwen2.5-32B · No SFT data



60.4

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

VALUE-MODEL-FREE (GRPO / DAPO)

- Single token credit assignment is not fully solving

VALUE-MODEL-BASED (VAPO)

- Value-based credit assignment (e.g. leaf node bonus)

Key insight: value-model-free methods assign identical advantage to every token in a trajectory.

Three Challenges in Value-Model-Based RL for Long-CoT

These bottlenecks prevent naive value-model approaches from scaling to complex reasoning.

01

VALUE MODEL BIAS

Bootstrapped value estimates compound errors over long trajectories. A single misestimated state value propagates through the entire return chain, causing instability and biased advantage computation.

02

HETEROGENEOUS LENGTHS

Responses range from dozens to thousands of tokens. A single λ (GAE) cannot optimize bias-variance tradeoff optimally — short responses need low λ (low variance) while long responses need high λ (low bias).

03

REWARD SIGNAL SPARSITY

Only the final `<eos>` token receives a reward signal. In trajectories with thousands of tokens, the vast majority of positions see zero reward, making learning extremely slow and noisy.

VAPO Framework: Integrated Solutions

Seven techniques mapped to three core challenges

VALUE MODEL BIAS

Value-Pretraining

Pre-trains value model before RL to reduce initial bias from random init

Decoupled GAE

Separates value estimation from policy GAE for robust advantage computation (from VC-PPO)

HETEROGENEOUS LENGTHS

Length-Adaptive GAE

Novel: dynamically adjusts λ per response length — short $\rightarrow$ low λ , long $\rightarrow$ high λ

Clip-Higher

Asymmetric clipping ratio from DAPO that encourages exploration beyond reference policy

REWARD SPARSITY

Token-level Loss

Normalizes loss at token level rather than sequence level (from DAPO)

Positive Example LM Loss

Self-imitation learning on correct trajectories (rewards past success)

Group-Sampling

Samples multiple responses per prompt for stable baseline estimation (from GRPO)

Length-Adaptive GAE

Dynamic λ Adjustment for Variable Response Lengths

Standard GAE

GAE(λ) computes advantage as an exponentially-weighted average of TD errors:

$$A_t = \delta_t + \lambda \gamma \delta_{t+1} + \lambda^2 \gamma^2 \delta_{t+2} + \dots$$

A fixed λ makes a single bias-variance tradeoff across all sequence lengths.

Length-Adaptive GAE (Ours)

λ is no longer a scalar — it varies with response length:

$$\lambda(L) = \min(1 - 1/L, \lambda_{\max})$$

- Short response (L=50): $\lambda \approx 0$ → low variance, fast convergence
- Long response (L=5000): $\lambda \approx 1$ → low bias, accurate credit assignment

This single change accounts for +15 points in ablation.

Bias-Variance Tradeoff Across Response Length

SHORT RESPONSE (L ≈ 50 tokens)

$\lambda \approx 0 \rightarrow A \approx \delta_t$ (single-step TD error)

Low variance ✓ Higher bias (short horizon)

↓ more tokens ↓

LONG RESPONSE (L ≈ 5 000 tokens)

$\lambda \approx 1 \rightarrow A \approx \sum \delta_t$ (Monte Carlo return)

Low bias ✓ Higher variance (long horizon)

KEY INSIGHT

A single λ across heterogeneous lengths is suboptimal. Length-Adaptive GAE automatically selects the optimal bias-variance tradeoff per sample, contributing +15 AIME points.

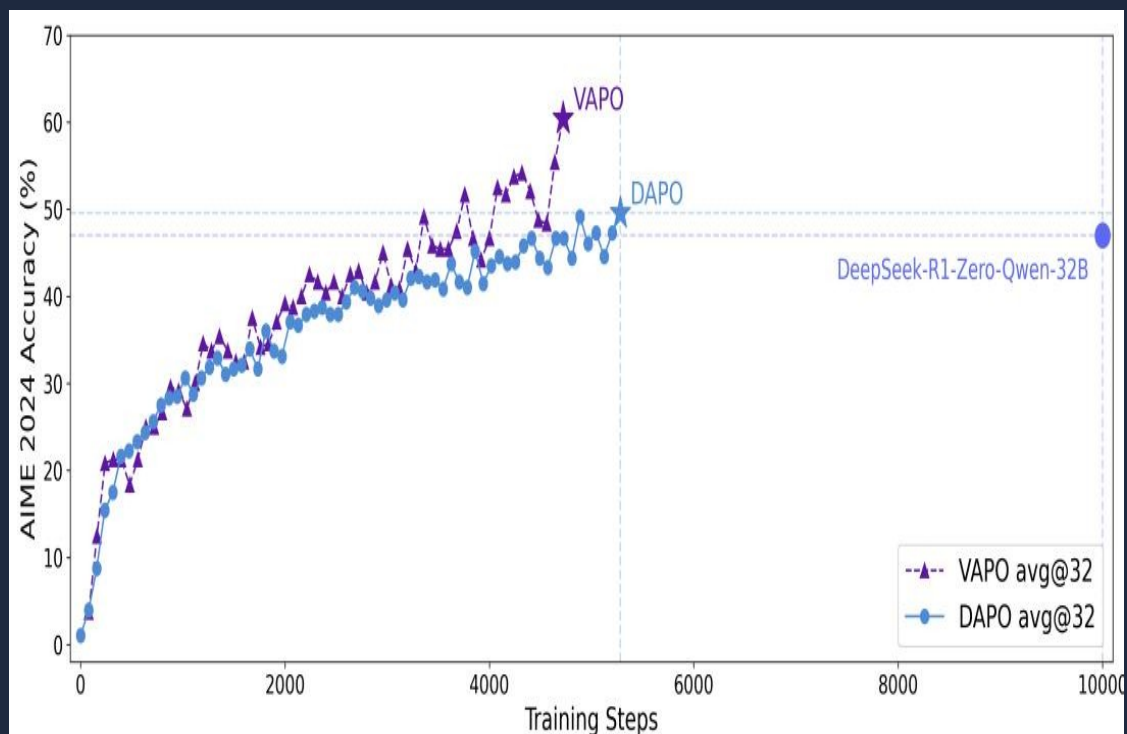
Ablation: removing Length-Adaptive GAE drops VAPO from 60 → 45 (−15 points)



VAPO Reaches SOTA in 5K Steps

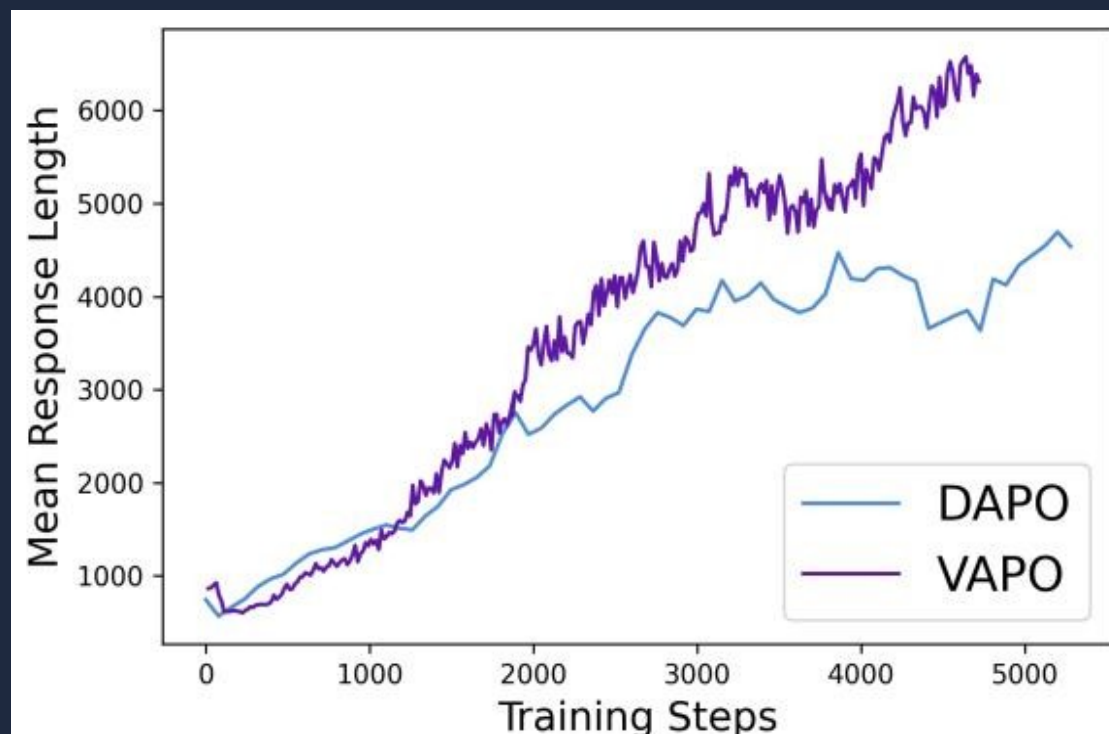
Half the training budget of DAPO — with consistent outperformance throughout

AIME 2024 Accuracy vs Training Steps (figure_2.jpg)



Source: figure_2.jpg

Mean Response Length vs Training Steps (figure_3.jpg)

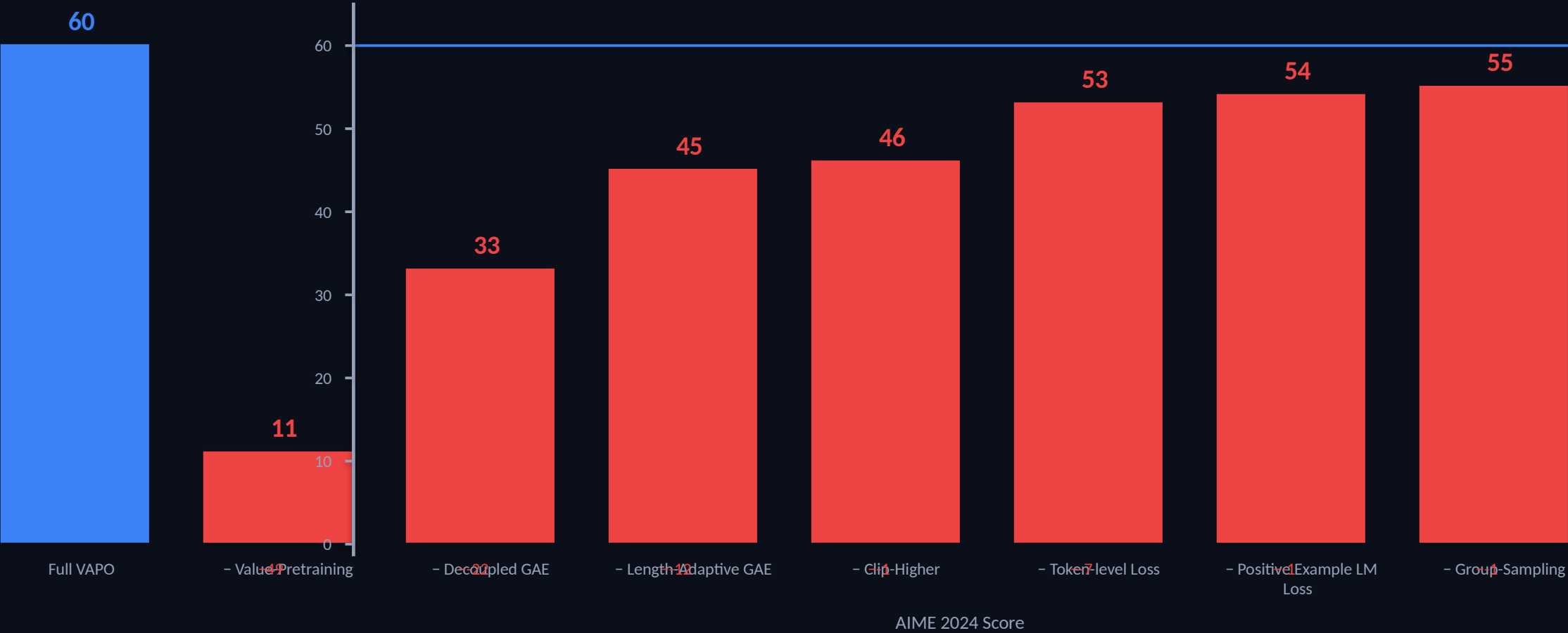


Source: figure_3.jpg

VAPO: SOTA in ~5 000 steps • DAPO: 50 @ ~5 500 steps • Zero training crashes across multiple runs

Every Component Matters

waterfall chart showing contribution of each VAPO component on AIME 2024



CRITICAL: Removing Value-Pretraining collapses VAPO from 60 → 11 (~49 points)

Value-Pretraining is the single most impactful component — without it, value-model-based RL barely works.

Key Takeaways & Implications

Value-Model-Based RL Is Back — With the Right Engineering

1

Value-model-based RL can outperform value-model-free

When the three core challenges (bias, heterogeneous lengths, reward sparsity) are systematically addressed, value models unlock superior reasoning performance.

2

Length-Adaptive GAE is critical for long-CoT tasks

Dynamic λ adjustment per response length optimally balances bias-variance tradeoff across heterogeneous sequence lengths (+15 points).

3

Value-Pretraining is the single most important component

Pre-training the value model on correct trajectories reduces initial bias from random initialization — removing it drops the score from 60 to 11 (~49 points).

4

VAPO is highly stable with zero training crashes

Across multiple independent runs, VAPO shows no training instabilities, reaching SOTA in ~5 000 steps (vs DAPO's ~5 500 steps for a lower score).

5

Implications for future reasoning model development

Long chain-of-thought reasoning demands token-level credit assignment. The VAPO framework provides a blueprint for scaling value-model-based RL to even longer and more complex reasoning chains.